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CHENNAI

Enhanced Feature Representation For Deep Learning-Based Classification of ECG Signals

(PhD Oral Viva Voce)

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Presentation Overview

1. Introduction
2. Literature Survey
3. Research Motivation
4. Research Contributions
5. Summary of Contributions
6. Limitations and Future Work
7. Publications
8. References

- **Contribution 1 — CHAF**
  Rethinking Activation Learning in CNN-Based ECG Classification
- **Contribution 2 — ECG-ResViT**
  Learning Beyond Local ECG Features
- **Contribution 3 — BlendNet**
  Beyond Temporal Modeling: Learning from Multiple ECG Representations

Introduction – Electrocardiogram (ECG)

What is an ECG?

- Records the heart's **electrical activity**
- Starts at the **sinus node** (natural pacemaker)
- **Non-invasive**, first-line cardiac test

What are its key waves & intervals?

- **P**: atrial depolarization
- **QRS**: ventricular depolarization
- **T**: ventricular repolarization
- **PR, QT, RR**: conduction and rhythm timing



What do clinicians look for?

- **Morphology**: shape, amplitude, width
- **Rhythm**: regularity of RR intervals
- **Intervals**: PR, QRS duration, ST-T changes

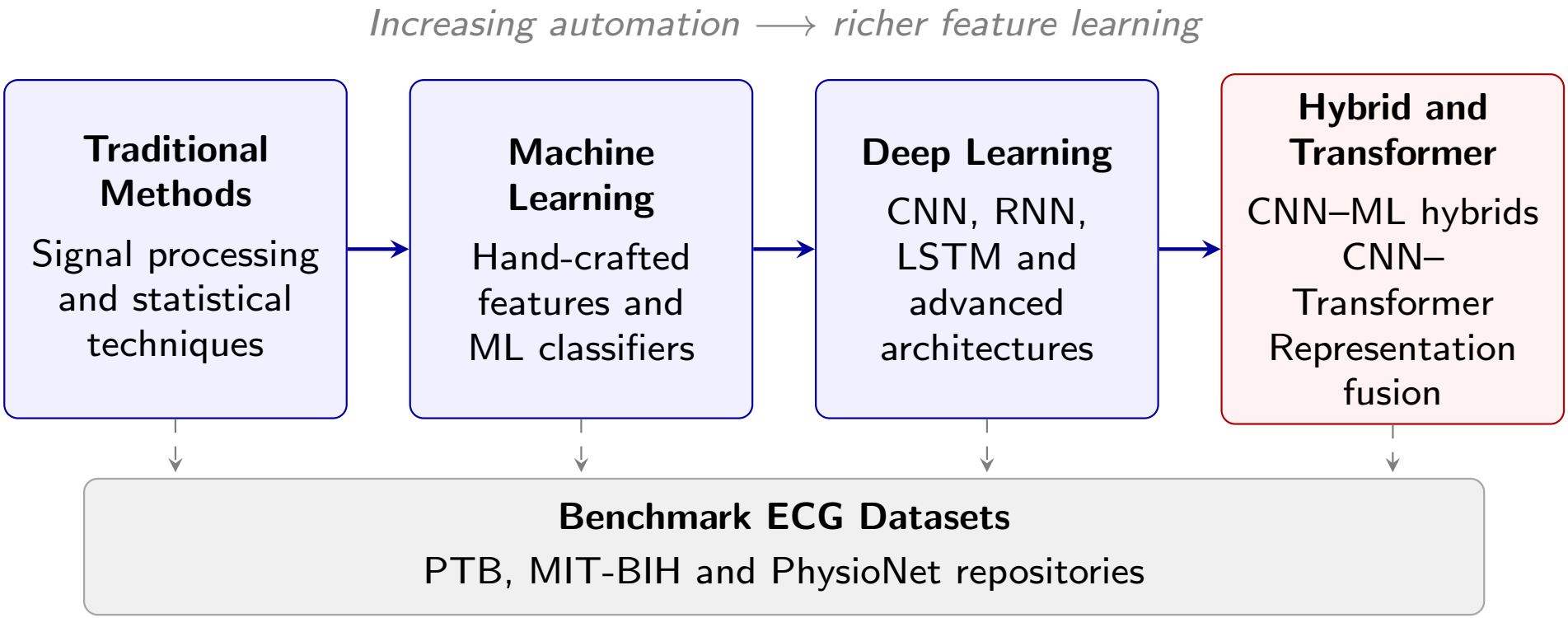
Why does it matter clinically?

- Reveals **arrhythmia**, **heart failure**, **ischemia**, **MI**
- Both **local waveform changes** and **beat-to-beat patterns** carry diagnostic information

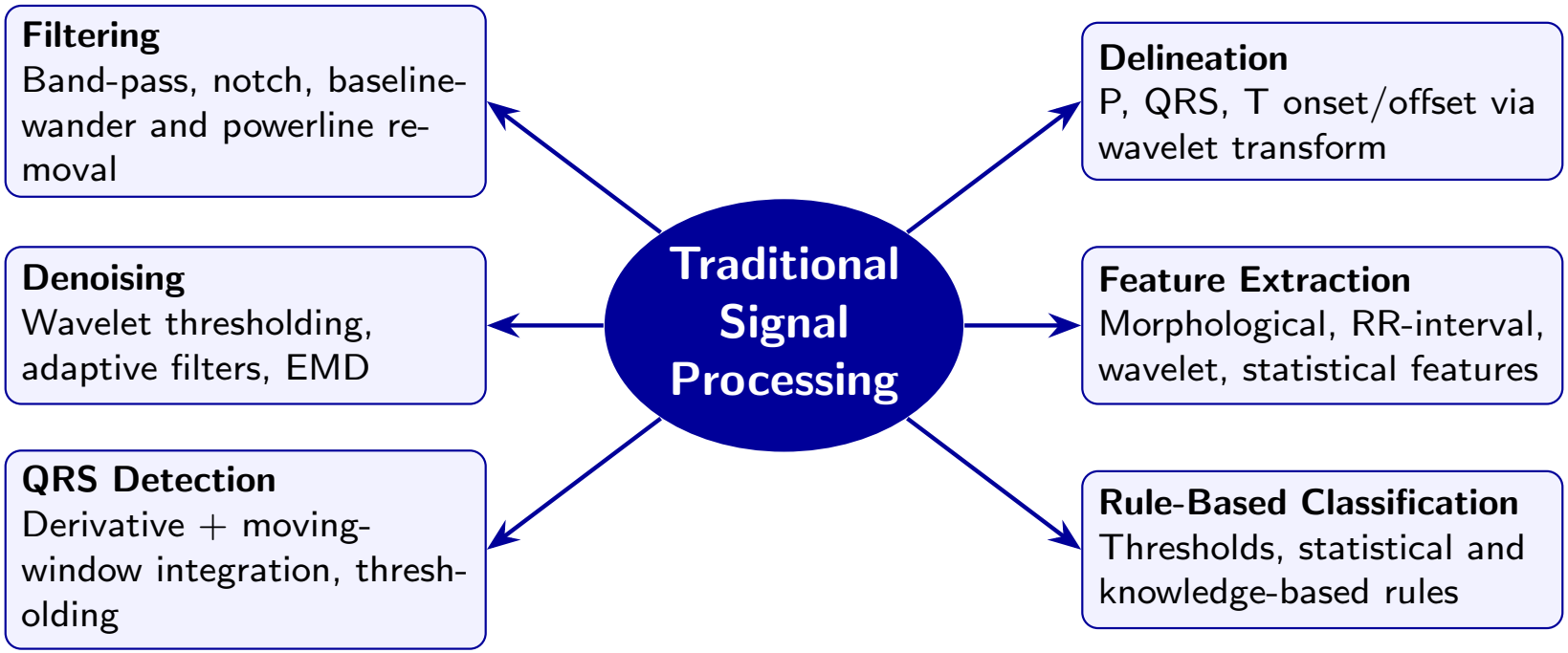
Global burden: cardiovascular diseases caused an estimated **19.8 million deaths in 2022**, about **32% of all global deaths** (WHO, 2025).

Literature Survey Overview

Automated ECG analysis literature was reviewed under the following categories:



Literature Survey: Traditional ECG Signal Processing



Enabled by: PhysioNet · MIT-BIH Arrhythmia DB · QT Database

Key Observation

Traditional analysis is a **multi-stage, hand-engineered pipeline**. Performance depends on each stage and on domain expertise, errors propagate between stages, and nothing is learned end to end. This motivates data-driven learning.

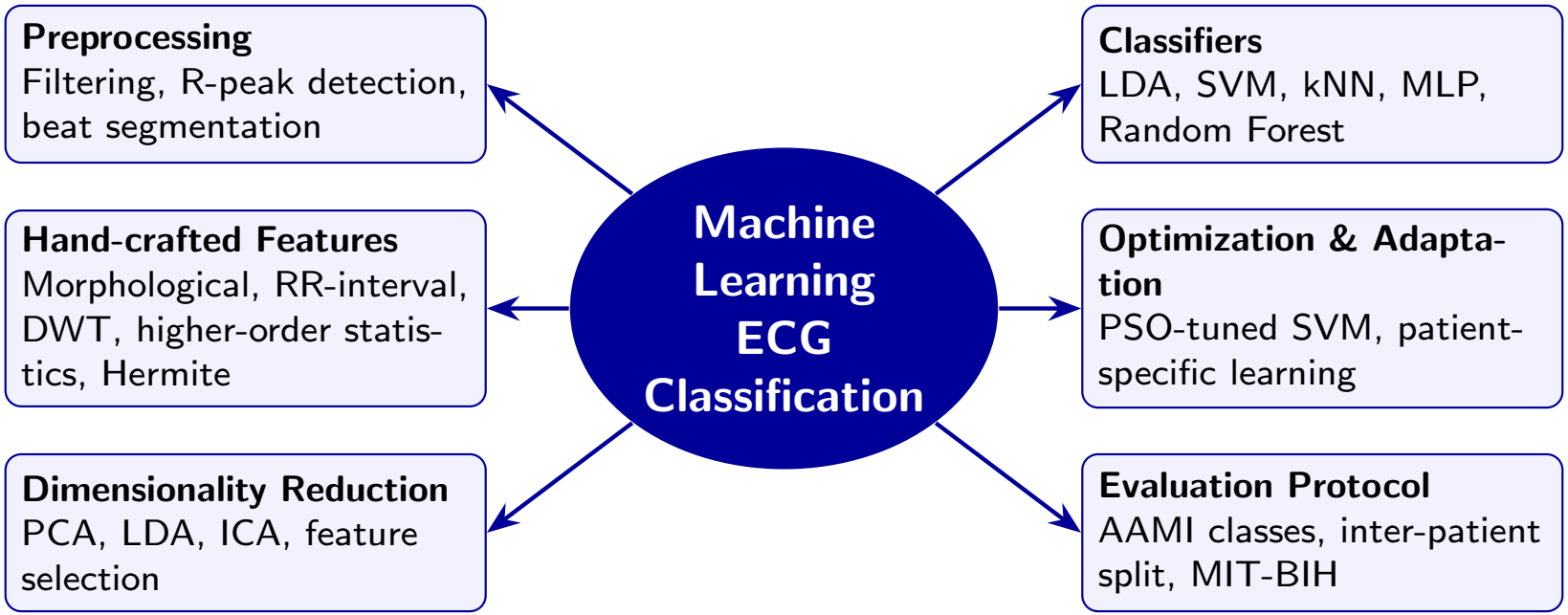
Key Works in Traditional Processing

Authors	Year	Venue	Key Contribution
Pan & Tompkins	1985	IEEE TBME	Real-time QRS detection by filtering and adaptive thresholds; the field's standard benchmark
Martínez et al.	2004	IEEE TBME	Wavelet-based delineation of P, QRS and T boundaries, validated on standard databases
Goldberger et al.	2000	Circulation	PhysioNet: the open repository behind most ECG research
Elgendi	2013	PLoS ONE	Fast knowledge-based QRS detector evaluated on 11 databases
Kalidas & Tamil	2017	IEEE BIBE	Stationary-wavelet real-time QRS detector
Satija et al.	2018	IEEE Rev. Biomed. Eng.	Review of signal processing and quality assessment for ECG
Makowski et al.	2021	Behav. Res. Methods	NeuroKit2: open-source toolbox for physiological signal processing

Takeaway

Classic methods (1985–2004) set the pipeline; recent work (2013–2021) refines individual stages and tools rather than learning end to end.

Literature Survey: Machine Learning-Based ECG Classification



Key Observation

ML improved ECG classification through **feature engineering and stronger classifiers**, but accuracy depended heavily on **feature quality and domain expertise**. Cross-patient (inter-patient) generalization remained a recognised challenge, and manual features limited scalability, which motivates automatic feature learning (deep learning).

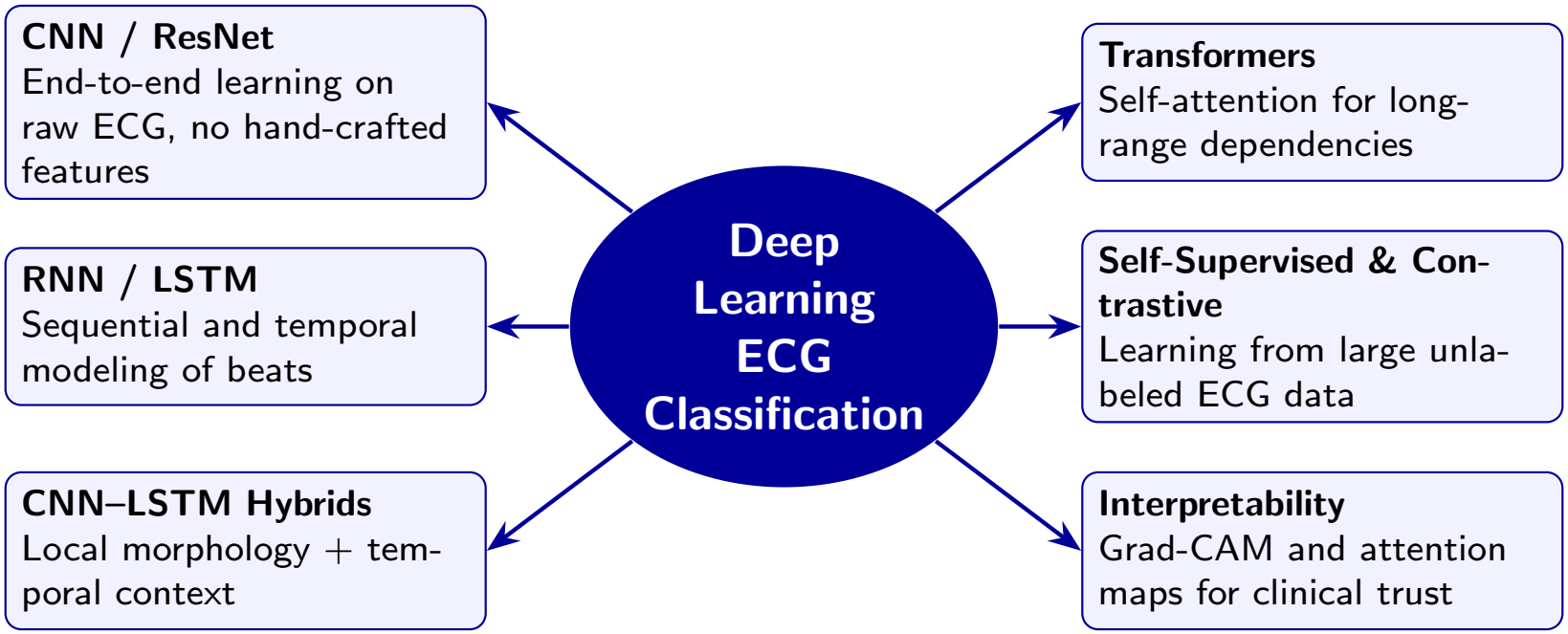
Key Works in Machine Learning-Based ECG Classification

Authors	Year	Venue	Key Contribution
de Chazal et al.	2004	IEEE TBME	Morphology + RR-interval features with LDA; set the inter-patient (AAMI) benchmark
Melgani & Bazi	2008	IEEE TITB	PSO-optimized SVM using morphological, wavelet and statistical features
Ince et al.	2009	IEEE TBME	Patient-specific MLP classifier needing few labeled beats
Llamedo & Martínez	2011	IEEE TBME	Feature selection driven by cross-database generalization
Martis et al.	2013	Biomed. Signal Process. Control	PCA/LDA/ICA with SVM and kNN; 98.91% with DWT + ICA + SVM
Luz et al.	2016	Comput. Methods Programs Biomed.	Survey of ECG heartbeat classification for arrhythmia detection
Mondéjar-Guerra et al.	2019	Biomed. Signal Process. Control	Ensemble of classifiers fusing temporal and morphological information

Takeaway

Progress came from better features and classifier design (2004–2019), yet every method still relied on **manually engineered features**. This is the gap that CNN-based learning, and later CHAF, addresses.

Literature Survey: Deep Learning-Based ECG Classification



Large-scale data: PhysioNet · PTB-XL · CPSC 2018 · MIT-BIH

Key Observation

Deep models **learn hierarchical ECG representations automatically** and clearly outperform traditional ML. Open challenges remain in **feature diversity, temporal dependency modeling, computational efficiency and representation fusion**.

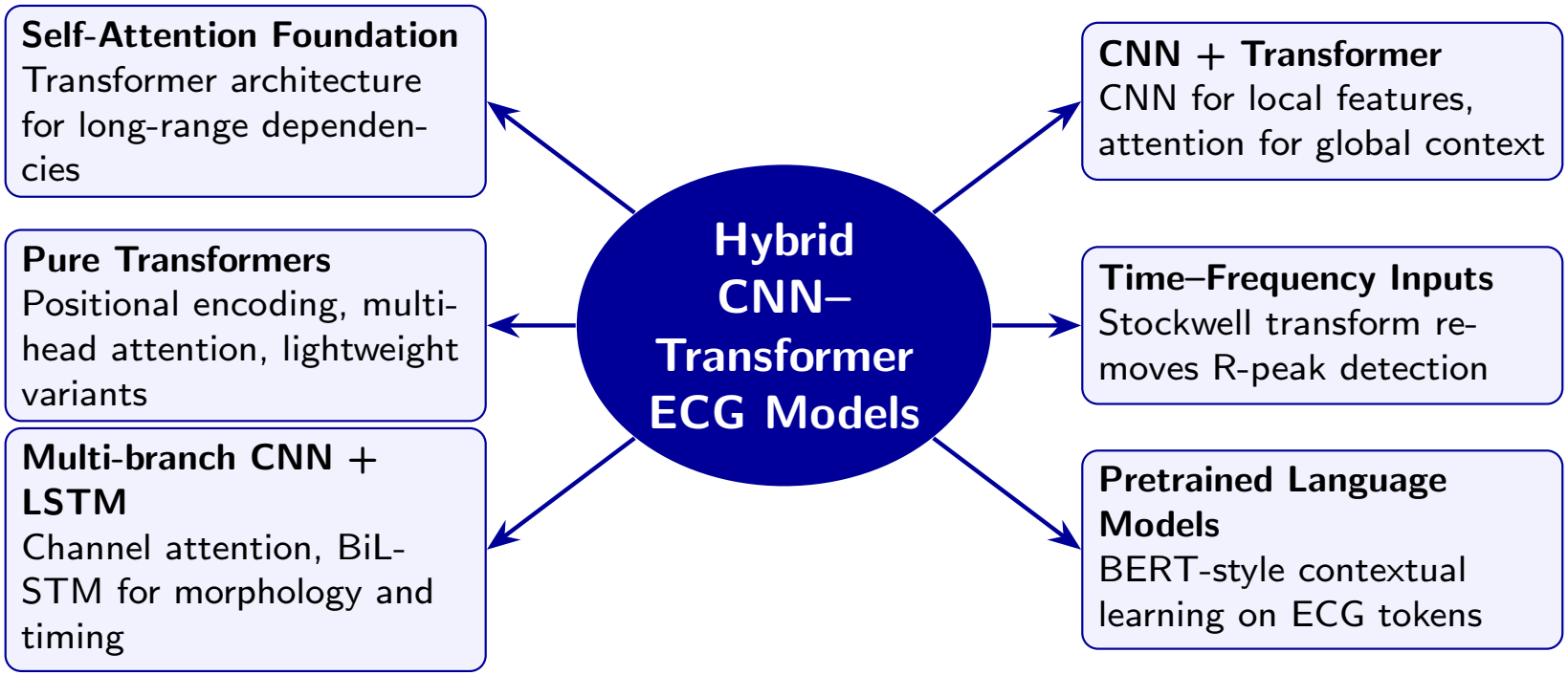
Key Works in Deep Learning-Based ECG Classification

Authors	Year	Venue	Key Contribution
Acharya et al.	2017	Comput. Biol. Med.	11-layer CNN on raw ECG segments for 5-class heartbeat classification
Hannun et al.	2019	Nature Medicine	34-layer CNN on 91,232 ECGs; cardiologist-level performance across 12 rhythms (F1 = 0.837)
Ribeiro et al.	2020	Nature Commun.	Residual CNN trained on a very large ECG corpus for six abnormalities
Strodthoff et al.	2021	IEEE JBHI	Benchmark of CNN, RNN and Transformers on PTB-XL
Hu et al.	2022	Comput. Biol. Med.	Pure Transformer for arrhythmia detection using self-attention
Lai et al.	2023	Nature Commun.	Self-supervised learning on large-scale 12-lead ECG for wearables
Sattar et al.	2024	Sensors	Comparison of CNN, LSTM and self-supervised autoencoders on digitized ECG

Takeaway

From CNNs on raw signals (2017) to Transformers and self-supervised learning (2022–2024), deep models gained accuracy. However, most still learn from a single ECG representation and use one activation family (typically ReLU-type) uniformly across layers. These open points motivate CHAF, ECG-ResViT and BlendNet.

Literature Survey: Hybrid CNN–Transformer ECG Methods



Benchmarks: PTB-XL · MIT-BIH · PhysioNet Challenge datasets

Key Observation

Transformer and hybrid models **improve global temporal dependency learning**. However, many remain **computationally expensive** and lack **efficient local–global feature integration**.

Key Works in Hybrid CNN–Transformer Methods

Authors	Year	Venue	Key Contribution
Vaswani et al.	2017	NeurIPS	Transformer and self-attention, the base of ECG Transformers
Strodthoff et al.	2021	IEEE JBHI	Benchmarked CNN, RNN and Transformers on PTB-XL
Meng et al.	2022	Artif. Intell. Med.	Lightweight Transformer with positional encoding for heartbeat classification
Liu et al.	2023	ISA Trans.	Multi-branch CNN with channel attention and BiLSTM
El-Ghaish & Eldele	2024	BSPC	ECGTransForm: bidirectional Transformer, adaptive positional encoding
Islam et al.	2024	BSPC	CAT-Net: CNN + channel attention + Transformer (local–global)
Kim et al.	2025	Sci. Rep.	Stockwell transform + CNN + Transformer, no R-peak detection
Musa et al.	2025	Sci. Rep.	ECGBert: 1D-CNN + BERT for local and contextual learning

BSPC: Biomed. Signal Process. Control

Takeaway

Hybrids (2022–2025) combine local and global learning, but many rely on heavy attention modules, 2D time–frequency inputs or extra branches, and few report complexity. ECG-ResViT targets this trade-off: residual-dilated CNN + compact-token Linformer ViT on native 1D ECG.

Research Motivation

What the survey found	Research need	Contribution
ECG CNNs typically use one fixed activation (mostly ReLU-type) in all layers, which may limit feature diversity	**Richer ECG feature representations**	**Contribution 1** CHAF
CNNs miss **long-range, multi-beat** dependencies; Transformers are **costly**	**Efficient long-range temporal modeling**	**Contribution 2** ECG-ResViT
Models learn from a **single ECG representation**; fusion methods are **complex**	**Complementary representations, fused simply**	**Contribution 3** BlendNet

Research Goal

Develop **accurate, efficient and robust** ECG classification frameworks through

Feature Diversity $\rightarrow$ **Temporal Modeling** $\rightarrow$ **Representation Fusion**

Research Objectives

Objective 1

To improve ECG feature diversity through layer-wise activation learning Convolution-based Heterogeneous Activation Facility (CHAF).

Objective 2

Develop a hybrid CNN–Transformer framework (ECG-ResViT) for joint local morphology learning and global temporal dependency modeling.

Objective 3

Develop a representation fusion framework (BlendNet) that integrates complementary ECG representations for robust cardiac diagnosis.

Overall Goal

Design accurate, efficient, and interpretable deep learning frameworks for automated ECG classification.

ECG Datasets Used in the Study

PTB

PTB Diagnostic

Subjects: 290 (549 rec.)
Age: 17–87
Rate: 1000 Hz
Leads: 15 (12 ECG + 3 Frank)
Status: multiple cardiac disorders
Labels: clinical diagnoses
Resolution: 16-bit

ARR

MIT-BIH Arrhythmia

Subjects: 47 (48 rec.)
Rate: 360 Hz
Leads: 2
Recording: 48 half-hour ECGs
Status: arrhythmia
Labels: ∼110K expert
Resolution: 11-bit

NSR

MIT-BIH NSR

Subjects: 18
Age: 20–50
Rate: 128 Hz
Leads: 2
Recording: long-term ECG
Status: no significant arrhythmia

CHF

BIDMC CHF

Subjects: 15
Age: 22–71
Rate: 250 Hz
Leads: 2
Recording: ∼20-hour ECG
Status: severe CHF (NYHA III–IV)
Resolution: 12-bit

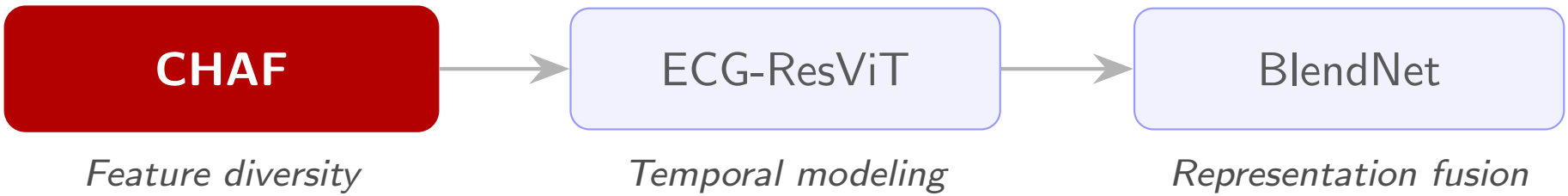
ARR–CHF–NSR: Objectives 2 & 3 (ECG-ResViT, BlendNet)

PTB + MIT-BIH: Objective 1 (CHAF)

Data used: Obj. 1: pre-processed beats (188 samples, 125 Hz), PTB 14,550 and MIT-BIH 109,444. **Obj. 2 & 3**: 162 recordings (ARR 96, NSR 36, CHF 30; counted per lead, 2 leads per subject), resampled to 128 Hz.

Contribution 1

Rethinking Activation Learning in CNN-Based ECG Classification



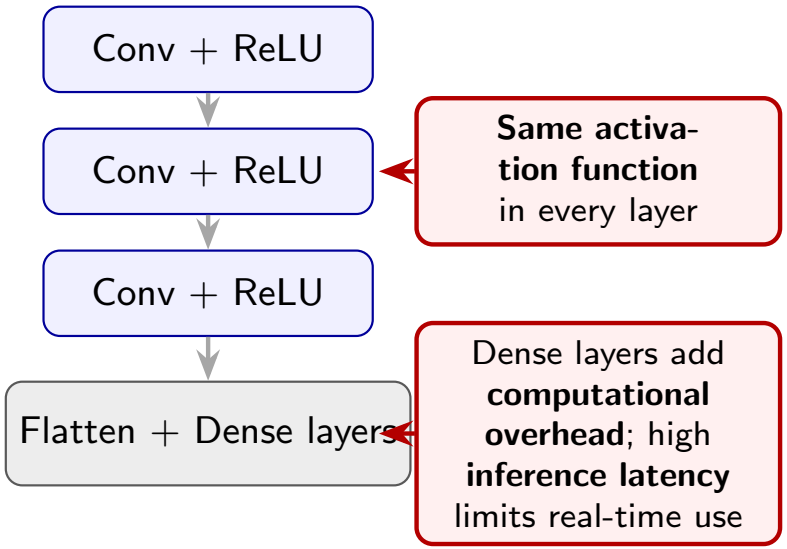
Question

Can **activation diversity** improve ECG feature learning without increasing model complexity?

Why Move Beyond Conventional CNNs?

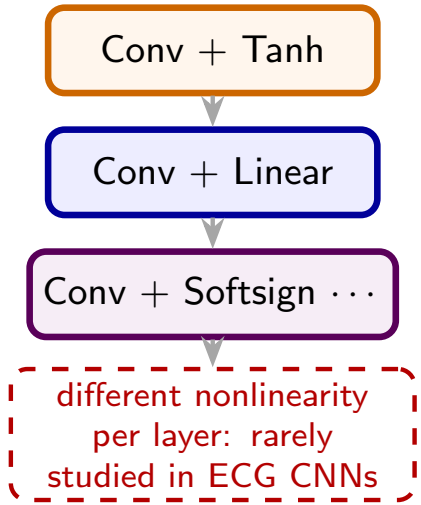
Existing deep CNN-based ECG models

+ high accuracy, hierarchical features learned automatically



vs

Layer-wise heterogeneous AFs

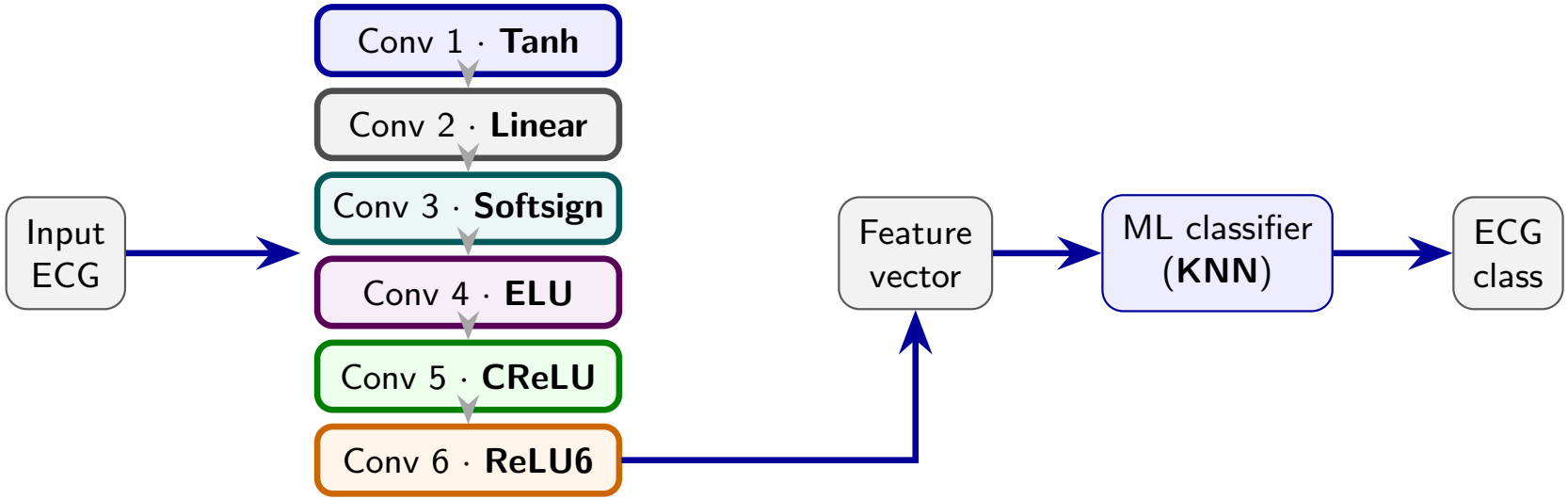


Research Gap

In ECG CNNs, the activation function is usually a fixed, network-wide design choice, although AFs differ in their nonlinear behaviour. The effect of assigning different fixed AFs to different convolution blocks on ECG feature representation and classification has received little attention

Proposed Solution: CHAF

Heterogeneous CNN (one activation per layer)



How the design was chosen (PTB and MIT-BIH)

1 Depth study
number of convolution blocks (ReLU baseline)

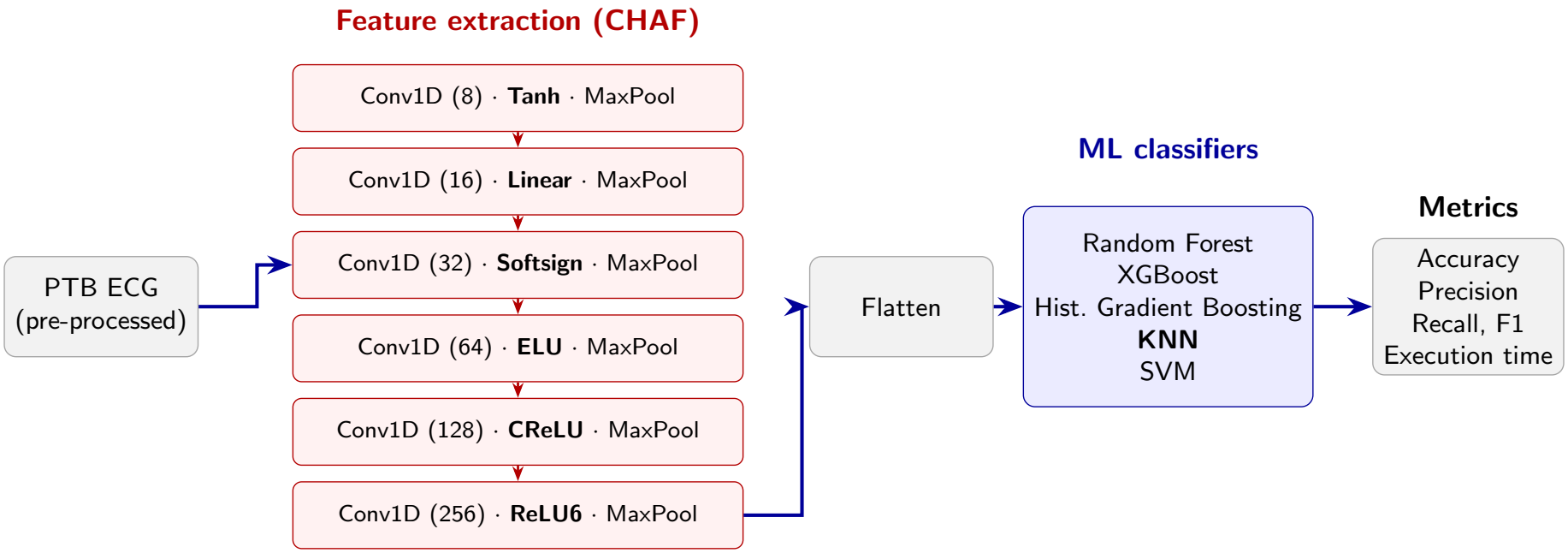
2 Activation ranking
compare different AFs and keep the top ones

3 Layer-wise combinations
heterogeneous AF per layer; best combination

Core Idea

Layer-wise heterogeneous activations learn **complementary ECG features**, and a **lightweight classifier replaces the dense layers**.

CHAF Architecture



Adapted from: S. Premanand and S. Narayanan, "Convolution-Based Heterogeneous Activation Facility for Effective Machine Learning of ECG Signals," *Computers, Materials & Continua*, vol. 77, no. 1, pp. 25–45, 2023. doi:10.32604/cmc.2023.042590

Optimal CNN Depth

Observation from Experiments:

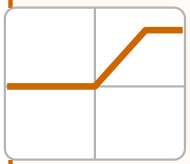
Blocks	Accuracy (%)	Time (s)
4	98.53	25.00
5	99.07	32.68
6	**99.18**	**47.12**
7	98.80	84.84
8	98.69	390.10

Key Insight:

- Accuracy improves up to **six convolution blocks** - PTB dataset
- Beyond six blocks: Accuracy and Computational cost
- Six blocks provide the optimal **bias–variance trade-off**

Why Heterogeneous Activation Functions?

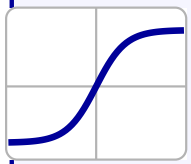
ReLU6 *QRS complexes*



$$f(x) = \min(\max(0, x), 6)$$

Captures sharp and high-amplitude **QRS complexes**

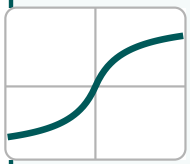
Tanh *P and T waves*



$$f(x) = \tanh(x)$$

Useful for modeling gradual variations in **P and T waves**

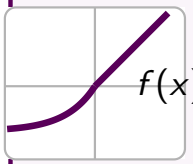
Softsign *subtle variations*



$$f(x) = \frac{x}{1 + |x|}$$

Helps capture **subtle waveform variations**

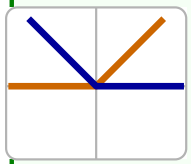
ELU *negative deflections*



$$f(x) = \begin{cases} x \\ \alpha(e^x - 1) \end{cases}$$

Captures **inverted or negative** ECG deflections

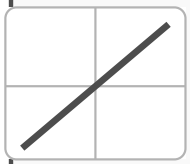
CReLU *both polarities*



$$f(x) = [\mathrm{ReLU}(x), \mathrm{ReLU}(-x)]$$

Captures both **positive and negative** signal patterns

Linear *raw morphology*



$$f(x) = x$$

Preserves **raw signal morphology**

A **heterogeneous mix** lets each function emphasise a different ECG characteristic, giving **richer and more diverse features**.

Activation Function Evaluation

Activation Function	Accuracy (%)	Time (s)
Swish	99.23	75.52
ELU	**99.33**	**72.36**
PReLU	99.16	77.77
SELU	99.04	52.84
CReLU	**98.92**	**67.36**
ReLU6	**99.16**	**66.19**
Leaky ReLU	99.27	68.32
Tanh	**99.46**	**46.91**
Sigmoid	72.14	45.48
GELU	99.34	54.41
Linear	**99.49**	**44.59**
Softsign	**99.31**	**49.20**

CNN–ML Architecture: Performance with AF

Performance of CNN–ML architecture for different activation functions (blocks fixed at 6)

Swish shown as a representative case; other activation functions follow the same format · Orange: highest accuracy / shortest time

Activation Function	ML Algorithm	Time (s)	Accuracy	Precision (0/1)	Recall (0/1)	F1 Score (0/1)
Swish	RF	10.58	98.80	99/99	97/99	98/99
	XGB	18.08	98.87	98/99	98/99	98/99
	HGB	22.78	**98.96**	98/99	98/99	98/99
	K-NN	**0.004**	98.85	98/99	97/99	98/99
	SVML	1.631	98.85	98/99	98/99	98/99
	SVMR	15.05	98.57	99/98	96/100	97/99
	SVMP	10.57	96.81	98/96	90/99	94/98

Observation

With Swish (6 blocks), accuracy ranges from 96.81% to 98.96%. HGB gave the highest accuracy (98.96%), while K-NN was the fastest (0.004 s).

CHAF–ML: Classifier Comparison

Performance comparison of CHAF–ML classifiers

CHAF–ML Algorithm	Time (s)	Accuracy (%)	Precision	Recall	F1-Score
CHAF-RF	2.90	99.36	99/99	99/99	100/99
CHAF-XGB	6.77	99.25	99/99	99/99	100/99
CHAF-HGB	16.58	99.25	99/99	99/99	99/99
CHAF-KNN	**0.008**	**99.55**	**99/100**	**99/100**	**100/100**
CHAF-SVML	0.706	99.41	100/99	99/100	100/100
CHAF-SVMR	15.54	99.38	99/99	99/100	100/99
CHAF-SVMP	2.09	99.43	100/99	100/99	100/99

Observation

Among the classifiers compared, CHAF-KNN gave the **highest accuracy (99.55%)** with the **shortest time (0.008 s)**, making it a suitable choice for lightweight ECG classification.

CHAF-KNN: Effect of Activation Function and K

Performance of CHAF with different activation functions across varying K

Model	$K = 3$			$K = 5$			$K = 7$			$K = 9$		
	Acc	F1	IT	Acc	F1	IT	Acc	F1	IT	Acc	F1	IT
CHAF-KNN tanh	99.3	99.5	0.008	99.2	99.5	0.006	99.4	99.6	0.006	99.0	99.3	0.006
CHAF-KNN linear	99.3	99.5	0.006	99.1	99.4	0.006	99.1	99.4	0.007	99.1	99.3	0.007
CHAF-KNN Softsign	99.2	99.4	0.007	99.2	99.4	0.011	99.4	99.5	0.008	99.3	99.5	0.012
CHAF-KNN elu	99.4	99.6	0.010	99.2	99.4	0.009	99.1	99.4	0.008	99.1	99.4	0.009
CHAF-KNN swish	99.1	99.4	0.021	99.2	99.5	0.013	99.4	99.6	0.021	99.0	99.3	0.022
CHAF-KNN relu	99.1	99.4	0.008	99.0	99.2	0.009	99.2	99.4	0.008	99.1	99.4	0.001
CHAF-KNN all AFs	99.1	99.3	0.008	**99.5**	**99.6**	**0.008**	99.4	99.6	0.007	99.2	99.4	0.009

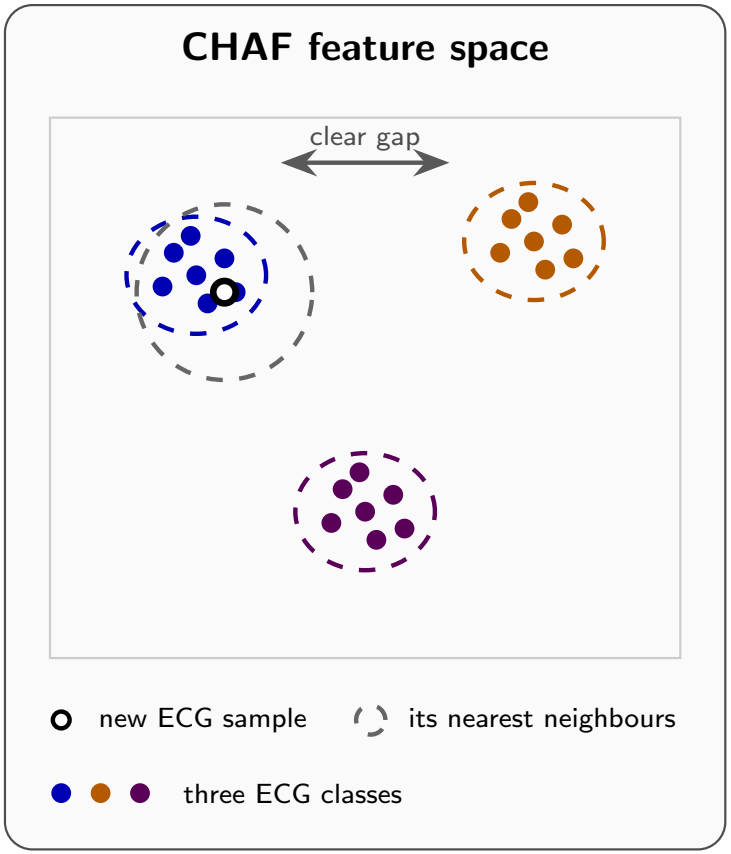
Key Observation

$k = 5$ gave a good **bias–variance balance**, with stable, high accuracy across the activation functions. Combining all AFs at $k = 5$ reached **99.5% accuracy** and **99.6% F1**. Minkowski distance suited the learned feature space well.

Acc = accuracy (%), F1 = F1-score (%), IT = inference time (s)

Why KNN with CHAF?

The **heterogeneous activation strategy of CHAF** learns highly **discriminative ECG feature representations** that improve **class separability**.



CHAF Features

highly discriminative ECG feature representations

Well-Separated Feature Space

similar classes form compact clusters; different classes are well separated

KNN Classification

distance-based and highly effective; no additional model training

High Accuracy

fast and computationally efficient inference

Experimental Validation

Table: Performance Evaluation of CHAF-KNN on PTB and MIT-BIH Datasets

S.No	Methodology	Data	Time (s)	Acc.	Prec.	Rec.	F1
1	CNN	PTB	47.12	99.18	98.62	99.03	98.82
2	CNN-KNN	PTB	0.006	99.31	99.12	99.42	99.26
3	CNN-XGBoost	PTB	1.23	99.38	99.38	99.28	99.32
4	**Proposed CHAF-KNN**	PTB	**0.008**	**99.55**	**99.55**	**99.80**	**99.68**
5	CNN	MIT-BIH	295.26	99.06	99.00	98.83	98.91
6	CNN-KNN	MIT-BIH	0.062	98.52	98.64	98.12	98.37
7	CNN-XGBoost	MIT-BIH	13.80	98.36	98.39	98.36	98.37
8	**Proposed CHAF-KNN**	MIT-BIH	**0.07**	**99.08**	**98.88**	**99.05**	**98.96**

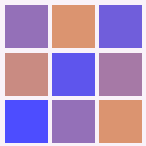
Key Observation

The proposed CHAF-KNN achieves the highest classification accuracy and F1-score on both PTB and MIT-BIH datasets while maintaining very low inference time.

Summary

Key Contributions

1. 
Layer-wise heterogeneous activation learning

2. 
Demonstrated that **activation diversity** improves ECG feature representation

3. 

Eliminated **dense-layer dependency**, reducing computational overhead

4. 

Achieved **99.55% accuracy** with **real-time inference** capability

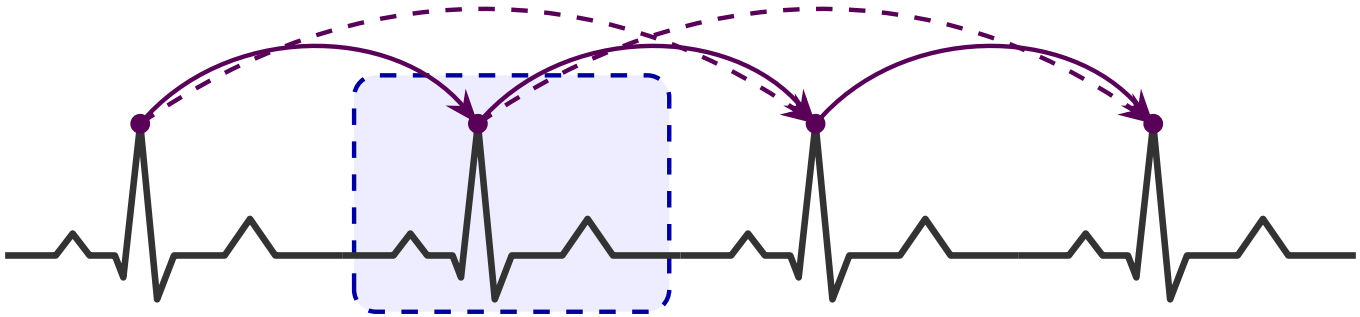
Takeaway

Activation diversity enables richer ECG feature learning **without increasing architectural complexity**.

Beyond Local ECG Features: Joint Local–Global ECG Learning

CHAF successfully improves **feature diversity** → *however* → ECG abnormalities often **span multiple cardiac cycles** → Rhythm disorders depend on **relationships between beats**

Global temporal dependencies across beats



Local ECG morphology

Key Question

How can we **simultaneously** learn **local ECG morphology** and **global temporal dependencies**?

Motivation for Objective 2: ECG-ResViT

Objective 2

Learning Beyond Local ECG Features



Question

How can we capture both **local ECG morphology** and **global temporal dependencies** within a single, efficient framework?

Why Move Beyond CHAF?

What CHAF achieved

✓ Improved **local feature diversity**

✓ Strong **local morphology** extraction

× Limited to short, local windows

What ECG diagnosis needs

Abnormalities span **multiple cardiac cycles**:

- Rhythm irregularities
- Inter-beat dependencies
- Long-term temporal evolution

What existing models offer

CNN: strong local, limited receptive field

LSTM / GRU: sequential, costly, hard to parallelize

Transformer / hybrids: global, but heavy and not tuned to 1D ECG

Key Question

How can we capture both **local ECG morphology** and **global temporal dependencies** within a single, **computationally efficient** framework?

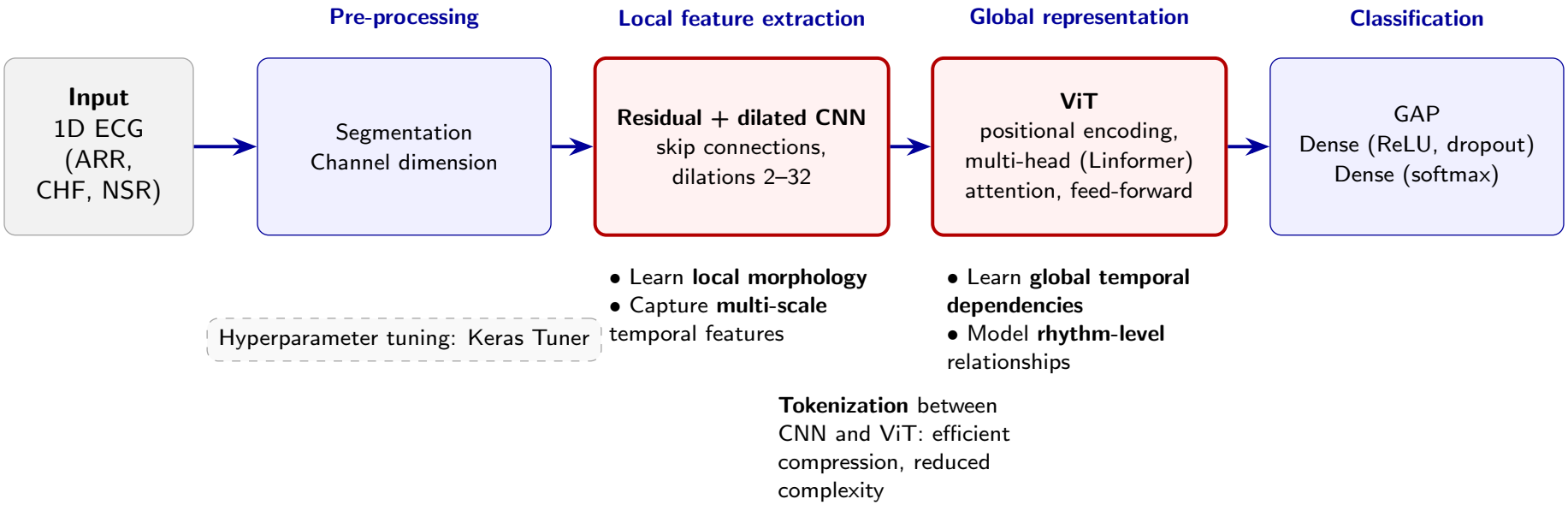
Research Gap

Gap	ECG-ResViT design choice
CNNs lack explicit global temporal modeling	**Residual-dilated CNN** for local and multi-scale features, then **ViT** for global dependencies
Transformers are computationally expensive	**Linformer** attention and compact **token** representation
Existing hybrids are not optimized for ECG-specific temporal characteristics	**Native 1D** processing, preserving the ECG structure

Research Question

Can we design a **computationally efficient hybrid architecture** that captures both **local ECG morphology** and **global temporal dependencies** while **preserving the native 1D ECG structure**?

Proposed Solution: ECG-ResViT



Core Idea

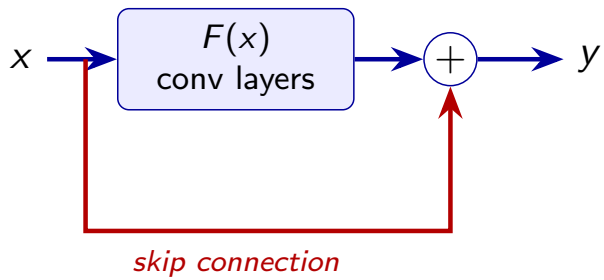
Combine the **local feature learning** of CNNs with the **global dependency modeling** of Transformers for **direct 1D ECG classification**.

Why Residual Learning?

Problem in deep CNNs

- **Feature degradation** in deeper layers
- **Vanishing gradients**
- Loss of important **ECG morphology**

Residual block



$$y = F(x) + x$$

ECG benefit

Preserve QRS morphology: waveform characteristics maintained

Deeper feature extraction

Stable training: better gradient propagation

Key Insight

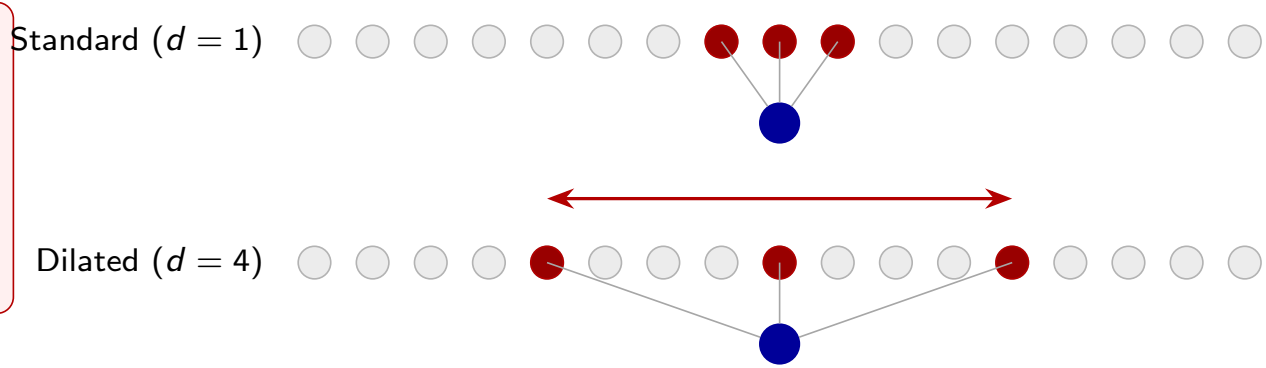
Residual learning **preserves critical ECG waveform information** while enabling **deeper, more discriminative** feature extraction.

Why Dilated Convolutions?

Problem

- ECG abnormalities span **multiple cardiac cycles**
- Standard convolutions see only a **limited temporal region**
- **Rhythm-level patterns** may be missed

Kernel sampling (size 3)



Same 3 weights, wider view

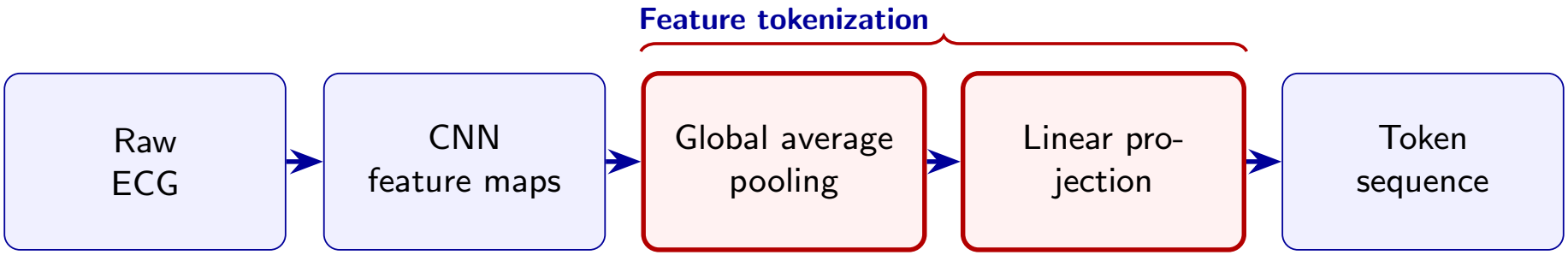
$$y(t) = \sum_{i=0}^{K-1} w(i)\, x(t - d \cdot i), \quad d \in \{2, 4, 8, 16, 32\}$$

Key Insight

Dilated convolutions **expand the temporal receptive field** and capture **multi-scale** ECG patterns **without adding parameters**.

From CNN Features to Transformer Tokens

Challenge: CNNs output **high-dimensional feature maps**, but Transformers expect **sequential tokens**, so a bridge is needed.



Benefit: compresses the representation, reduces computational complexity and produces **Transformer-compatible tokens**.

Key Insight

Tokenization turns rich CNN features into **compact sequential representations** suitable for global dependency learning.

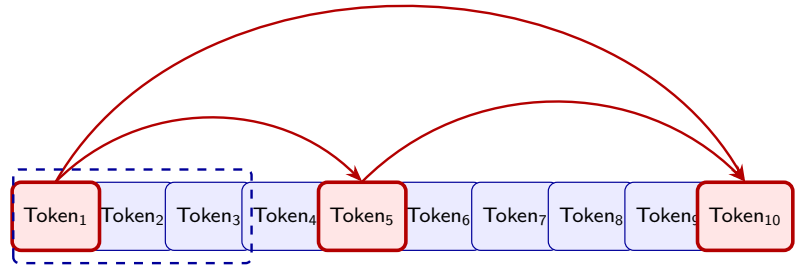
Why Vision Transformer?

Problem

- Diagnosis depends on relationships **across multiple cardiac cycles**
- CNNs focus on **local waveform** characteristics
- **Long-range** dependencies are hard to capture

Self-attention across beats

ViT attention: any beat to any beat



CNN window: local only

Benefit

- **Global** temporal relationships
- Dependencies across **distant segments**
- **Rhythm-level** characteristics

Key Insight

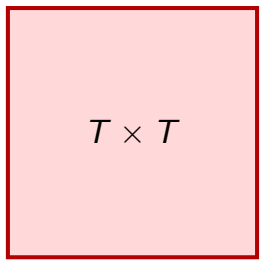
ViT complements the CNN by learning **global temporal dependencies** beyond the receptive field of convolutional layers.

Why Linformer?

Problem

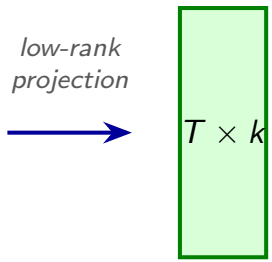
- Standard self-attention relates **all token pairs**
- Cost grows **quadratically**: $O(T^2)$
- **Long ECG sequences** need large memory and computation

Standard attention



$O(T^2)$

Linformer attention



$O(Tk),\ k \ll T$

Benefit

- **Reduced memory** consumption
- **Faster** training and inference
- Suitable for **long ECG** recordings

Key Insight

Linformer preserves the benefits of self-attention while **significantly reducing computational complexity**.

State-of-the-Art Comparison

Study	Methodology	Key Finding	Acc. (%)
Khadirnaikar (2020)	AlexNet + SVM	Hybrid deep learning and machine learning framework for ECG classification	96.77
Sabeenian (2023)	Transfer Learning + Wavelet Scattering	ResNet18 achieved superior performance with reduced computational cost	98.81
Kafal (2023)	CNN vs GoogLeNet	Lightweight CNN outperformed deeper GoogLeNet architecture	91.20
Sankar Ganesh (2024)	CWT + Spectrograms	SqueezeNet achieved highest performance using image-based ECG representation	98.70
Kolhar (2024)	CQ-NSGT + Dual Branch Fusion	Hybrid CNN architecture achieved high validation and test accuracy	98.82
Proposed	**Hybrid CNN-ViT**	**Residual-dilated CNN with ViT for local-global ECG feature learning**	**99.13**

Research Gap

Most existing studies rely on CNN-based image representations. Hybrid CNN-ViT architectures capable of simultaneously learning local morphology and global dependencies remain largely unexplored.

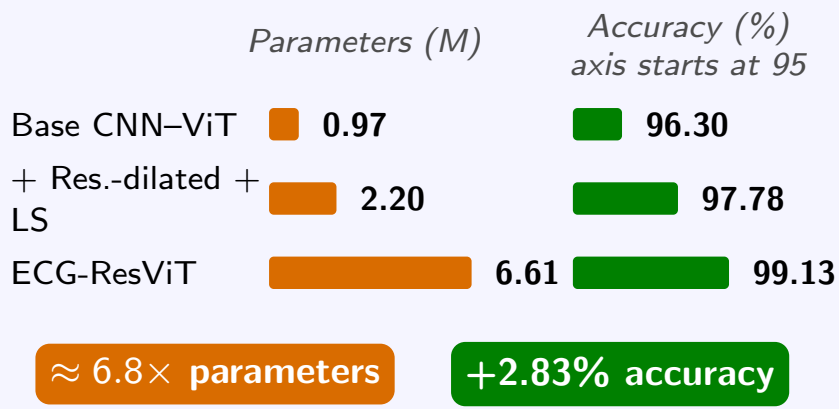
ECG-ResViT: Design Choice and Computational Cost



Why an encoder-only ViT?

- **One label per segment**: an encoder is enough, no decoder needed
- **Compact tokens**: attention runs on CNN-derived tokens ($T = 10$), not raw samples
- **Linformer-style** low-rank attention keeps attention cost small
- **Local + global**: CNN morphology plus self-attention on native 1D ECG

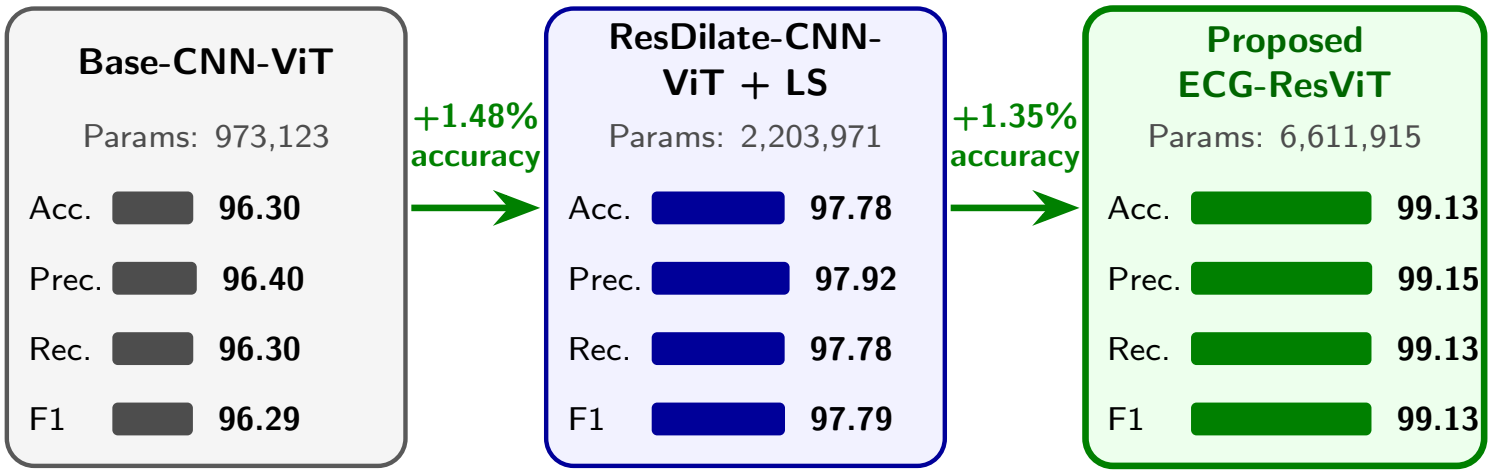
Accuracy vs. model size



Conclusion
ECG-ResViT is an **accuracy-oriented hybrid**, not a demonstrated lightweight model.

Not yet measured
FLOPs, inference latency and memory; no controlled comparison with a temporal Transformer

Ablation Study on Different Architectures



Key Observation

Residual-dilated convolutions, label smoothing, and hyperparameter tuning raise accuracy from **96.30%** to **99.13%** (+2.83% accuracy), with precision, recall, and F1 improving at each step. Parameters grow from 0.97 M to 6.61 M.

Summary of ECG-ResViT and Motivation for BlendNet

ECG-ResViT achieves

- ✓ Local **morphology** learning (residual-dilated CNN)
- ✓ Global **temporal dependencies** (ViT)
- ✓ Efficient attention (**Linformer**)
- ✓ Native **1D** ECG processing
- ✓ **99.13%** accuracy on ARR–CHF–NSR

What is still missing

Learns from a **single ECG representation**, so there is only **one feature space**.

Spectral and structural views are **never combined**.

Motivation for Objective 3: BlendNet

Objective 3

Beyond Temporal Modeling: Learning from Multiple ECG Representations



Question

Can **complementary ECG representations** be fused to generate richer and more discriminative cardiac features?

Motivation and Research Gap

So far: single view

CHAF: better features via heterogeneous activations

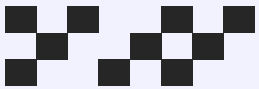
ECG-ResViT: better learning via CNN–Transformer

both learn from **one** *ECG representation*

Complementary views



Scalogram (time–frequency): spectral information



Binary map (structural): shape and boundaries

Gap

1 Independent processing: spectral and structural information are processed separately

2 Complex fusion: existing fusion relies on complex architectures

3 Interpretability: simple, interpretable fusion is underexplored

Research Question

Can **simple blending** of complementary ECG representations improve classification performance?

Comparison with Existing ECG Fusion Approaches

Method	Fusion Strategy	Dataset	Accuracy	Key Limitation
Bimodal CNN (2023)	Grayscale + Scalogram	Chapman–Shaoxing 12-lead (4 classes)	95.08% (Lead II)	Dual backbone architecture (high complexity)
Magnitude–Phase Fusion (2024)	Scalogram + Phasogram	MIT-BIH Arrhythmia	≈98.5%	Multiple fusion schemes required
GAF-FusionNet (2024)	Attention-based Multimodal Fusion	ECG200 / ECG5000 / MIT-BIH	94.5% / 96.9% / 99.6%	Additional attention module complexity
SIFT-CNN (2024)	Handcrafted Features + CNN	MIT-BIH	99.78%	Dependency on manual feature extraction
BlendNet (Proposed)	**Alpha-Controlled Feature Blending**	**ARR / CHF / NSR (162 recordings)**	**99.21% ± 1.05% (CV)**	**Lightweight and Interpretable Fusion**

Metrics as reported by each paper; datasets and class definitions differ, so values are indicative and not directly comparable.

Key Observation

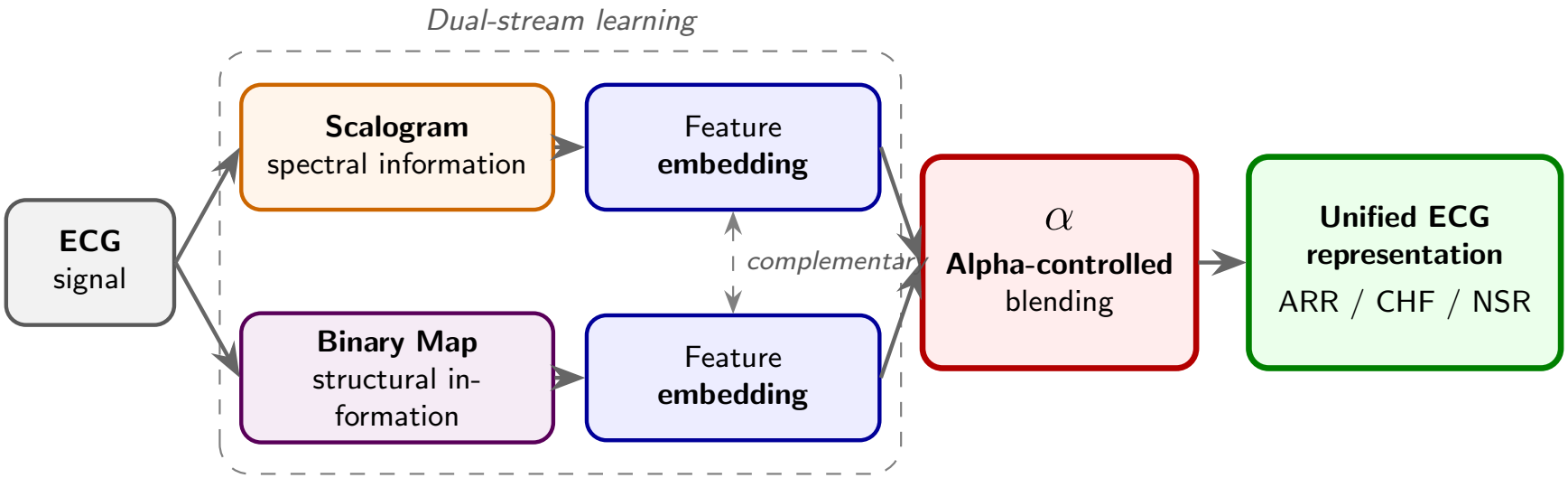
- Existing fusion methods rely on **complex fusion architectures**
- Limited focus on **simple and interpretable** representation integration

Research Need

- **Lightweight** fusion framework for complementary ECG representations
- Controlled and **explainable** feature blending mechanism

Proposed Solution: BlendNet

What if a single ECG representation is not enough?

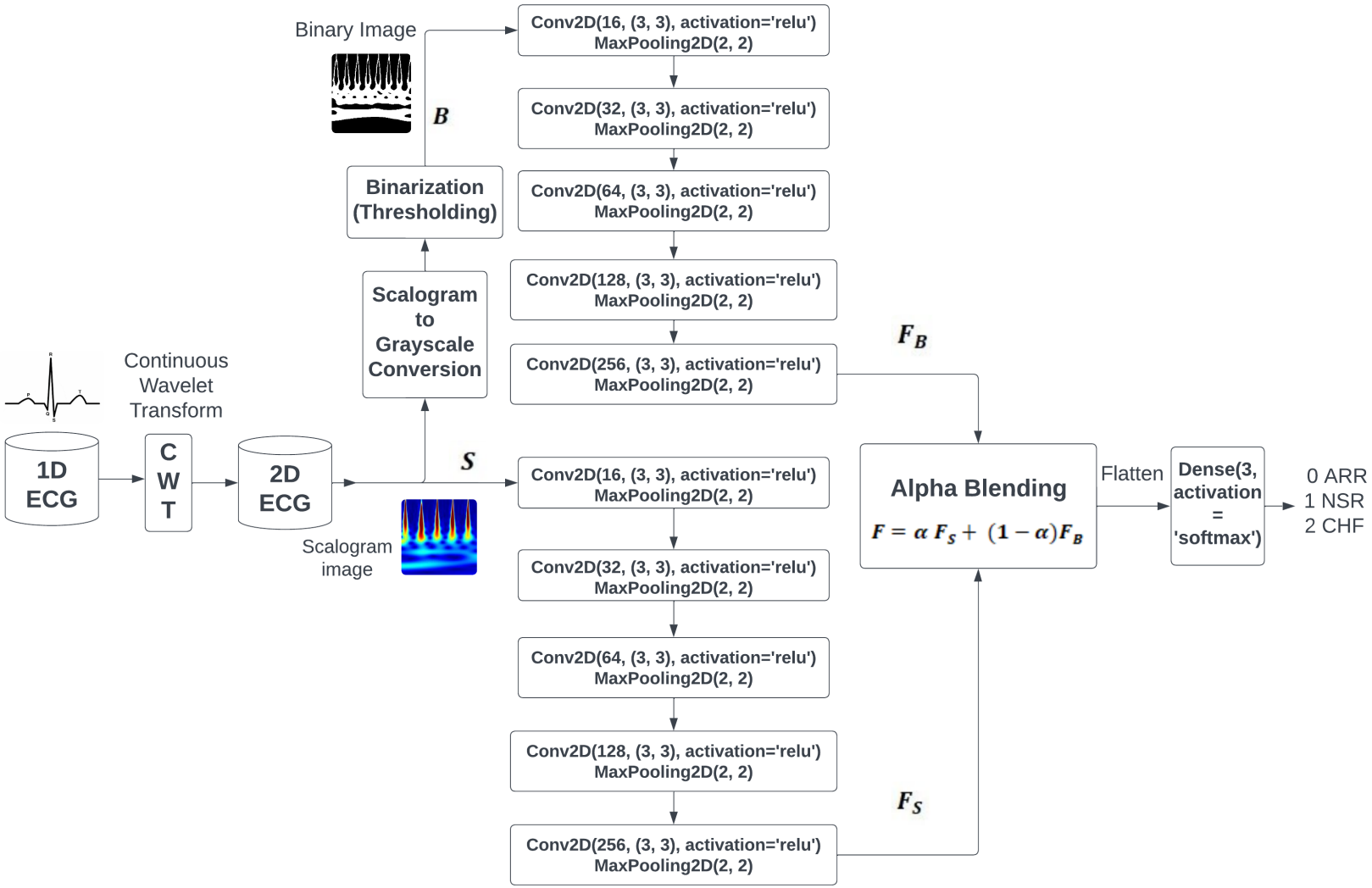


Key Innovation

Feature-level blend-ing, not image-level fusion

Lightweight, interpretable and effective

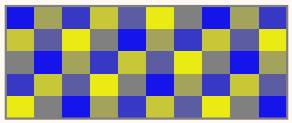
BlendNet Architecture



ECG Signal-to-Image Representations

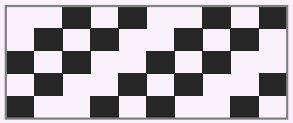
ECG signal

Scalogram (CWT)



- Spectral Dynamics
- Multi-Scale Information
- Frequency Evolution
- Energy Distribution

Binary Representation



- Structural Morphology
- Dominant Boundaries
- Shape Characteristics
- Pattern Geometry

Different Representations $\rightarrow$ Different Insights

Complementary Information $\neq$ Redundant Information

Alpha-Controlled Feature Blending

Scalogram captures **spectral and temporal dynamics**

Binary captures **structural and morphological patterns**

Research Question

What is the optimal balance between spectral and structural information for ECG classification?

$$F = \alpha F_s + (1 - \alpha) F_b$$

F_s: spectral embedding

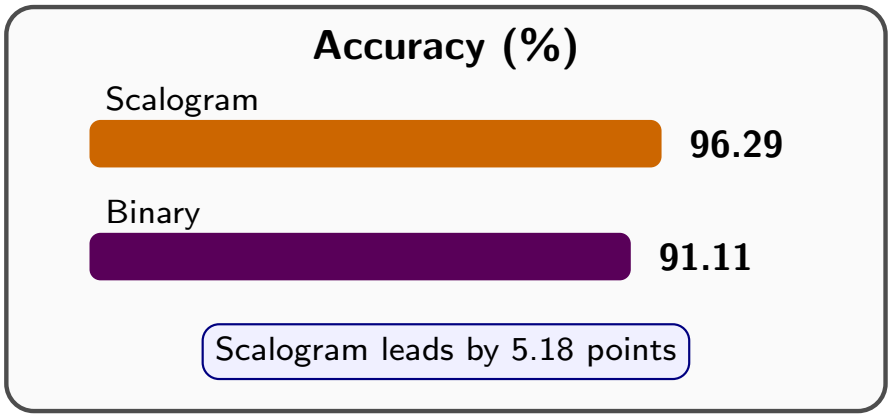
F_b: structural embedding

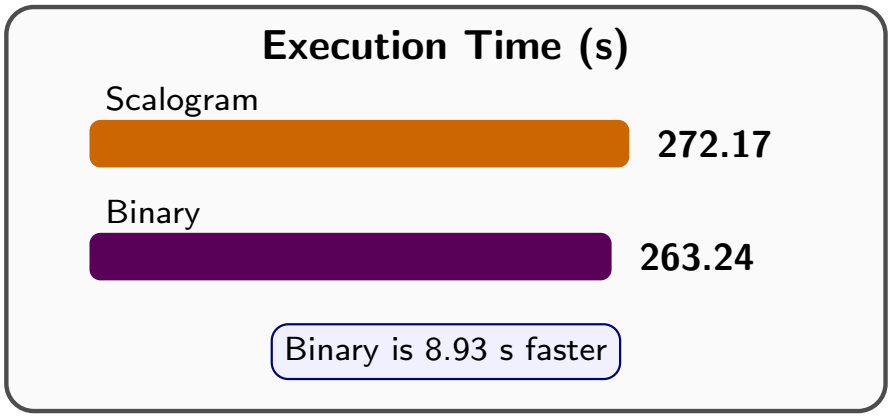
α: control coefficient

Interpretation of α

$\alpha = 0$	$0 < \alpha < 1$	$\alpha = 1$
Structural-only representation	Hybrid representation learning	Spectral-only representation

Baseline Performance: Single-Representation Models





Key Observation

Scalogram > Binary Representation

Representation Choice Influences Classification Performance

Spectral Features ≠ Structural Features

Complementary Information Exists Across Representations

Research Insight

Single Representation → Incomplete Feature Learning

Motivation

Can Complementary ECG Information be Integrated?

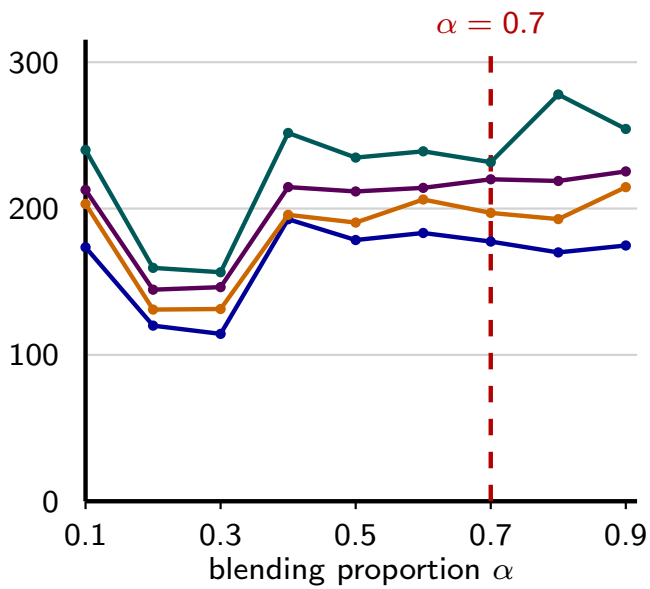
BlendNet Performance for Different Blending Ratios

Accuracy (%) by train:test split

α	60:40	70:30	80:20	90:10
0.1	96.11	95.18	92.22	96.66
0.2	94.07	95.18	94.44	93.33
0.3	92.59	94.07	93.88	95.55
0.4	95.92	97.40	95.00	94.44
0.5	93.33	94.81	91.66	96.66
0.6	94.07	95.55	93.33	96.66
0.7	**100.00**	**100.00**	**100.00**	**100.00**
0.8	95.55	90.70	89.44	96.66
0.9	93.33	92.22	92.77	93.33

Execution time (s)

line colours match the train:test splits



Key Observation

At $\alpha = 0.7$, BlendNet reaches **100% accuracy in all four train:test splits** (neighbouring $\alpha = 0.6$ and 0.8: 89–97%).

At $\alpha = 0.7$, the **60:40 split is the fastest (177.43 s)**; shortest run times overall: $\alpha = 0.2$–0.3.

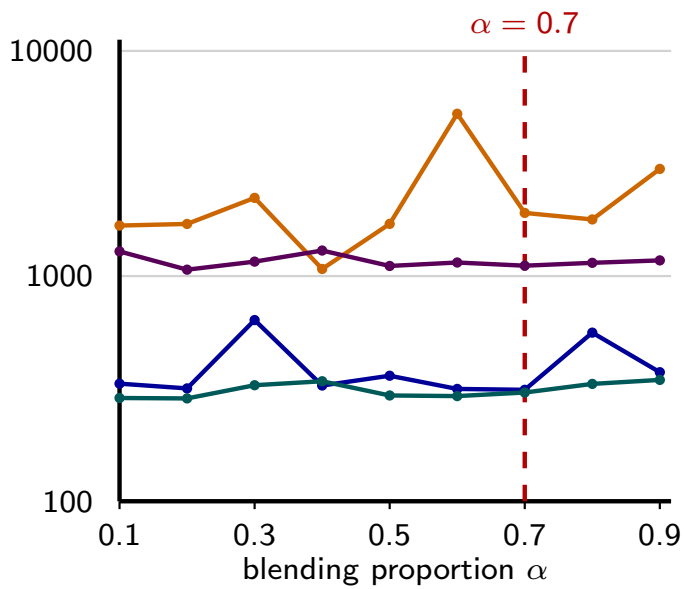
BlendNet Performance Across CNN Backbones

Accuracy (%) by backbone

α	InceptionV3	ResNet152V2	DenseNet169	MobileNetV2
0.1	90.37	89.25	94.07	93.70
0.2	83.33	90.74	91.85	90.74
0.3	85.92	91.11	89.62	91.85
0.4	85.18	88.51	91.48	91.48
0.5	87.03	91.85	89.99	86.29
0.6	89.99	93.70	90.74	89.25
0.7	**97.47**	**97.77**	**99.62**	**99.25**
0.8	88.88	92.22	88.51	91.48
0.9	88.51	91.85	92.22	92.22

Execution time (s, log scale)

line colours match the backbones



Key Observation

$\alpha = 0.7$ gives the **best accuracy on all four backbones** (97.47–99.62%); **DenseNet169 is highest (99.62%)**.

MobileNetV2 is the fastest (about 290–345 s; 304.03 s at $\alpha = 0.7$), while ResNet152V2 is the slowest (up to 5260.9 s).

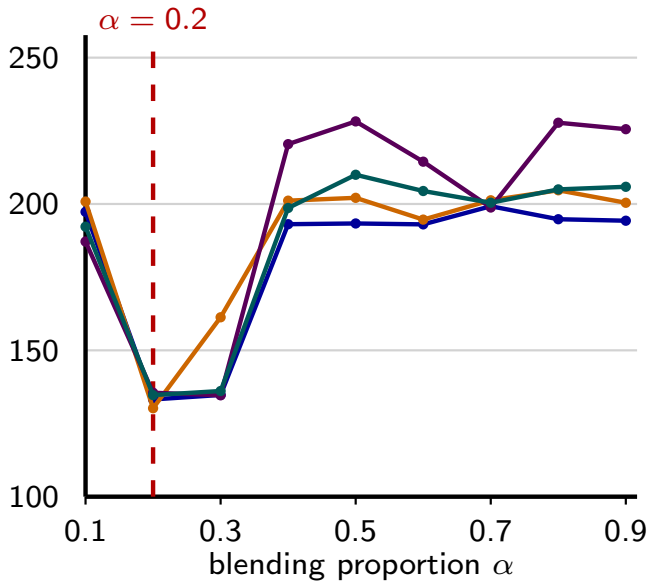
BlendNet Performance with Different ML Classifiers

Accuracy (%) by classifier

α	SVM	RF	KNN	XGBoost
0.1	97.03	96.66	96.29	96.29
0.2	**98.88**	**100.00**	**98.88**	**99.44**
0.3	97.22	97.22	97.22	97.22
0.4	95.18	95.18	95.18	94.81
0.5	96.66	96.66	96.66	97.03
0.6	94.81	94.81	94.81	94.44
0.7	92.22	92.59	91.85	92.22
0.8	93.33	93.33	93.33	93.33
0.9	93.33	92.96	93.33	93.33

Execution time (s)

line colours match the classifiers

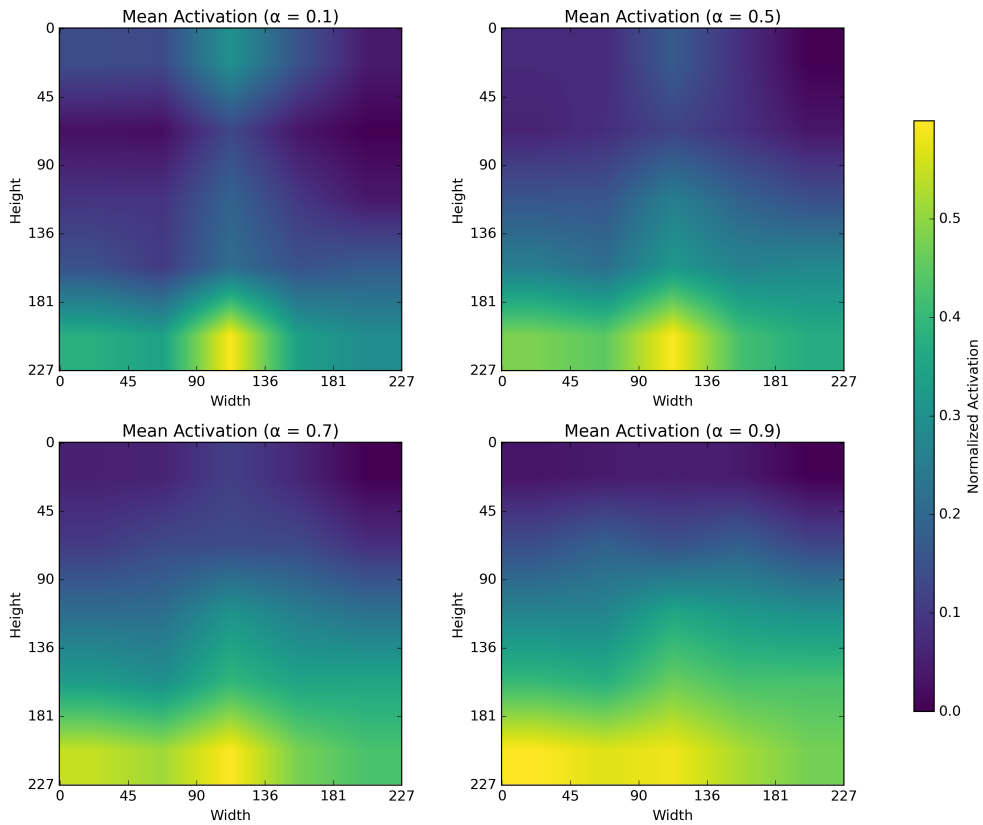


Key Observation

$\alpha = 0.2$ gives the **best accuracy for every classifier** (98.88–100%); **BlendNet-RF reaches 100%**.

RF is also the fastest at $\alpha = 0.2$ (**130.22 s**); accuracy falls to about 92% at $\alpha = 0.7$, so the best α depends on the classifier.

Mean Activation Heatmap of different α values



How to read it

Mean feature activation over the 227×227 input. **Yellow** = strong, **dark** = weak. $\alpha = 0.1$: structural-dominant; $\alpha = 0.9$: spectral-dominant.

What changes with α

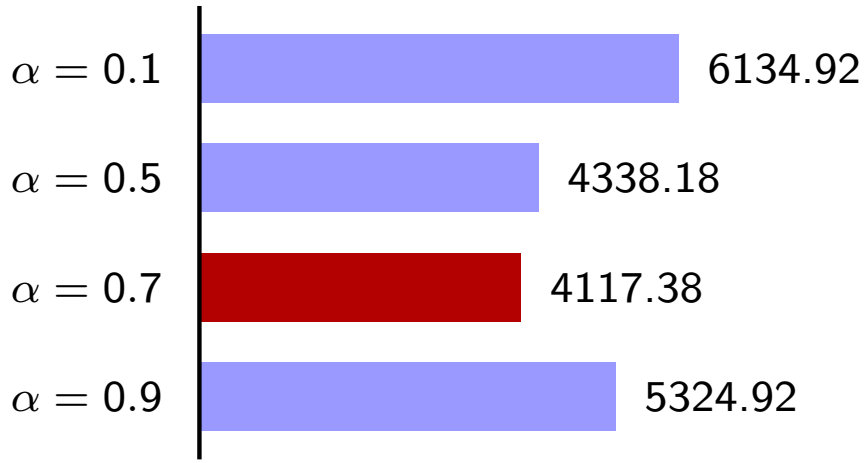
0.1: narrow central band at the bottom. **0.5/0.7**: band widens, upper region gains activation. **0.9**: spreads across the full bottom width.

Key Observation

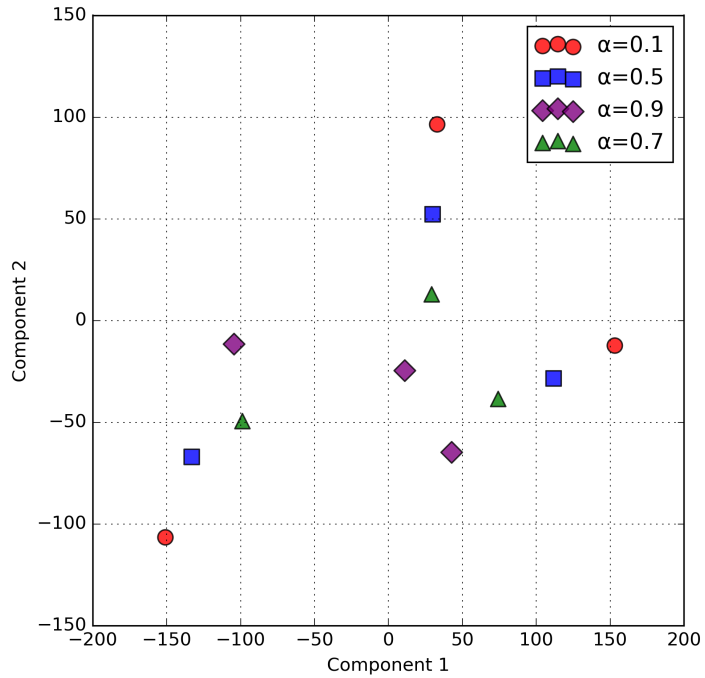
α redistributes where the network responds. Intermediate weights combine focused structural and broader spectral responses, consistent with best accuracy at $\alpha = 0.7$.

Feature Space Analysis using t-SNE

Within-Cluster Sum of Squares (WCSS)



Lower WCSS = tighter clusters. Each point in the t-SNE is a feature embedding projected to 2D.



t-SNE of feature embeddings for different α.

Key Observation

Lowest WCSS at $\alpha = 0.7$ (4117.38): the tightest clusters, agreeing with the best accuracy. Intermediate α beats both extremes (0.1, 0.9).

Comparison with State-of-the-Art Approaches

Reference	Architecture	Performance
Li et al. (2022)	Slantlet Transform + SVM Classifier	AUC: 99.25%
Yoon and Kang (2023)	SIFT + CNN (5-fold Cross Validation)	Accuracy: 99.78%
Proposed	**BlendNet with CNN-type Dense Layer ($\alpha = 0.7$)**	**CV: 99.21% $\pm$ 1.05%**

Key Observation

BlendNet **consistently outperforms** existing approaches, achieving the **best classification accuracy** using both **CNN-based dense layers** and **Random Forest** classifiers.

Summary

Idea

Two Representations, One Decision

Spectral + Structural Intelligence

Method

Alpha-Controlled Feature Fusion

Interpretable Learning Process

Gain

Feature Diversity Without Complexity

Improved Class Separability

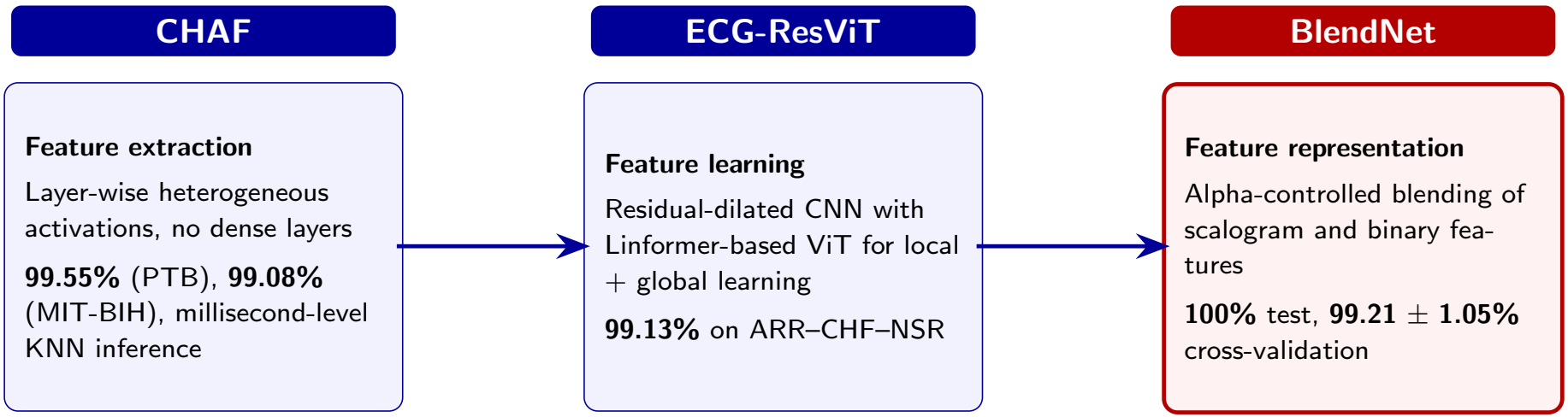
Outlook

Lightweight Yet Effective

Towards Real-Time ECG Classification

$$F = \alpha F_s + (1 - \alpha) F_b$$

Summary of Contributions



Overall Impact

Robust, accurate and computationally efficient automated ECG classification

Limitations and Future Work

CHAF

Limitation

Activation functions chosen empirically; depends on R-peak detection; single lead (Lead II)

Future work

- **Automated activation search** (NAS, meta-learning)
- **Multi-lead** extension with lead-aware activations

ECG-ResViT

Limitation

Self-attention cost on long recordings; FLOPs, latency and memory not measured

Future work

- **Report FLOPs, latency, memory**
- **Edge-efficient models**: pruning, quantization, distillation

BlendNet

Limitation

Blending weight α fixed (grid search) and fixed binarisation threshold

Future work

- **Adaptive blending**: learn α with attention
- **Threshold robustness** across signal quality

Cross-cutting future directions: validation, deployment and beyond

1. Robust validation

- **Patient-level split**, external datasets
- **Cross-device** testing
- **Noise robustness** (NST, graded SNR)

2. Clinical translation

- **Prospective validation** with cardiologists
- **Sensitivity, specificity, calibration**
- **Regulatory** requirements

3. Transparency

- **Explainable AI**: Grad-CAM, attention maps
- **KNN beat explanations** (CHAF)
- **Federated learning**

4. Broader scope

- **12-lead ECG** with attention over leads
- **Other biosignals**: EEG, PPG

Outlook: from benchmark results towards **validated, explainable and deployable** ECG decision support

Publications

1. S. Premanand and S. Narayanan, "BlendNet: A Blending-Based Convolutional Neural Network for Effective Deep Learning of Electrocardiogram Signals," *Frontiers in Artificial Intelligence*, vol. 8, 2025. doi:10.3389/frai.2025.1625637.
(SCI Indexed, Q1, Impact Factor: 4.7)

2. S. Premanand and S. Narayanan, "ECG-ResViT: A Hybrid CNN-ViT Model for Efficient ECG Signal Classification," *The European Physical Journal Special Topics*, vol. 234, pp. 4483–4496, 2025. doi:10.1140/epjs/s11734-025-01781-y.
(SCI Indexed, Q2, Impact Factor: 2.4)

3. S. Premanand and S. Narayanan, "Convolution-Based Heterogeneous Activation Facility for Effective Machine Learning of ECG Signals," *Computers, Materials & Continua*, vol. 77, no. 1, 2023. doi:10.32604/cmc.2023.042590.
(SCI Indexed, Q2, Impact Factor: 3.1)

4. S. Premanand and S. Narayanan, "A Tree Based Machine Learning Approach for PTB Diagnostic Dataset," *Journal of Physics: Conference Series*, vol. 2115, no. 1, p. 012042, 2021. doi:10.1088/1742-6596/2115/1/012042.
(Scopus Indexed Conference Publication)

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[1] J. Pan and W. J. Tompkins, "A Real-Time QRS Detection Algorithm," *IEEE Trans. Biomed. Eng.*, vol. 32, no. 3, pp. 230–236, 1985.

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Thank You!